

ST JOHNS, AZ, USA

WMO: 723754

Lat: 34.518N

Lon: 109.379W

Elev: 1747

StdP: 82.02

Time Zone: -7.00 (W07)

Period: 99-19

WBAN: 93027

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-12.0	-9.5	-21.9	0.7	2.3	-19.3	0.8	3.1	14.2	8.6	12.5	8.4	0.9	170	0.546

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.1	35.5	14.8	33.9	14.6	32.7	14.4	18.1	26.0	17.6	25.9	17.1	25.5	5.1	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.3	14.4	19.5	15.6	13.8	19.6	14.7	12.9	19.8	58.2	25.7	56.4	25.7	54.7	25.2	24.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.6	11.0	9.3	-17.1	37.5	4.2	1.5	-20.1	38.5	-22.6	39.4	-24.9	40.2	-28.0	41.3		
(5)			-17.8	19.6	3.9	1.2	-20.6	20.4	-22.9	21.1	-25.1	21.8	-27.9	22.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.0	1.7	4.3	8.1	12.2	16.9	22.9	24.8	23.1	19.8	13.2	6.7	1.6		
	DBStd	8.90	4.28	4.09	3.95	3.57	3.93	2.64	2.12	1.99	2.62	3.71	4.14	4.24		
	HDD10.0	910	257	163	82	19	3	0	0	0	0	14	112	260		
	HDD18.3	2528	515	394	317	184	72	2	0	1	14	162	350	518		
	CDD10.0	2006	1	2	23	86	218	386	459	407	296	114	12	1		
	CDD18.3	581	0	0	0	1	29	138	201	149	59	3	0	0		
	CDH23.3	7813	0	0	18	96	712	2261	2193	1523	862	146	3	0		
	CDH26.7	3480	0	0	1	8	235	1169	1111	654	283	20	0	0		
(14) Wind	WSAvg	3.3	2.6	3.4	3.9	4.6	4.3	4.0	3.3	2.9	2.9	2.9	2.8	2.7		
(15) Precipitation	PrecAvg	275	17	13	19	11	10	11	44	55	35	27	15	19		
	PrecMax	484	83	45	65	57	66	96	102	205	123	100	51	58		
	PrecMin	128	0	0	0	0	0	0	0	4	0	0	0	0		
	PrecStd	76	16	12	15	12	13	17	28	35	28	24	12	16		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.8	21.1	25.7	27.8	33.7	36.8	38.0	34.9	32.6	28.9	23.5	17.8		
		MCWB	6.5	7.3	9.5	10.4	14.0	14.3	16.1	15.7	14.7	12.8	8.9	6.0		
	2%	DB	15.1	18.7	22.9	26.0	31.2	35.3	36.1	33.3	31.2	27.0	21.4	15.1		
		MCWB	5.1	6.4	8.1	9.4	12.4	13.9	15.5	15.5	14.2	12.0	8.2	5.4		
	5%	DB	12.7	16.2	21.0	24.1	29.1	33.9	34.1	32.1	29.8	25.0	19.0	12.6		
		MCWB	4.3	5.3	7.2	8.6	11.2	13.6	15.5	15.4	13.9	10.9	7.5	4.4		
	10%	DB	10.6	13.6	18.7	22.6	27.5	32.7	32.5	30.6	28.2	23.0	16.8	10.4		
		MCWB	3.5	4.4	6.4	8.1	11.0	13.2	15.5	15.5	13.6	10.1	6.7	3.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.2	8.8	10.4	11.3	14.7	16.7	19.0	18.9	17.3	15.0	10.7	8.1		
		MCDB	14.7	17.0	22.7	24.8	31.6	26.2	26.4	25.8	25.0	23.3	18.6	13.7		
	2%	WB	6.5	7.3	8.8	10.1	13.2	15.6	18.3	18.1	16.5	13.5	9.4	6.5		
		MCDB	12.5	16.2	21.0	23.8	28.1	28.8	26.2	25.8	24.3	22.2	18.0	12.8		
	5%	WB	5.1	6.2	7.8	9.3	12.2	15.0	17.8	17.7	15.9	12.4	8.3	5.2		
		MCDB	11.2	14.3	19.2	22.8	26.4	29.6	26.1	25.8	24.0	21.8	17.4	11.5		
	10%	WB	3.9	5.1	6.8	8.5	11.2	14.2	17.3	17.2	15.3	11.3	7.2	3.9		
		MCDB	9.6	12.7	17.3	21.4	25.5	29.5	25.8	25.3	23.7	21.0	16.1	9.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.6	16.1	17.7	17.9	17.9	18.8	15.1	14.7	16.3	17.4	17.5	15.1		
		MCDBR	19.2	21.0	22.1	20.8	20.1	19.9	17.8	17.1	18.3	20.4	21.5	19.1		
		MCWBR	11.3	11.2	10.9	9.7	8.5	7.5	5.2	4.8	6.2	9.1	11.3	11.2		
		MCDBR	16.5	17.9	19.7	19.1	18.4	17.6	13.8	13.5	13.7	16.0	18.3	16.4		
	5% WB	MCWBR	10.1	9.7	9.8	9.0	8.1	6.7	3.8	4.0	4.8	7.3	10.0	10.0		
(40) Clear-Sky Solar Irradiance	taub		0.239	0.245	0.269	0.283	0.296	0.306	0.359	0.346	0.307	0.273	0.247	0.238		
	taud		2.649	2.642	2.548	2.495	2.503	2.472	2.432	2.432	2.533	2.599	2.636	2.652		
	Ebn at Noon		1002	1024	1013	1004	986	970	917	927	956	974	981	983		
	Edh at Noon		71	80	96	107	107	111	114	112	96	82	71	67		
(44) All-Sky Solar Radiation	RadAvg		3.26	4.21	5.69	7.07	7.85	8.12	6.76	6.29	5.78	4.79	3.67	2.90		
	RadStd		0.29	0.35	0.35	0.34	0.51	0.43	0.41	0.26	0.22	0.38	0.21	0.19		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (7 neighbors)	+0.42	N/A	N/A	N/A	N/A	N/A	N/A	-100	+73	+36	

Nomenclature: See separate page